

torch.nn.functional.cosine_similarity#

https://pytorch.org/docs/stable/generated/torch.nn.functional.cosine_similarity.html

Description:

Returns cosine similarity between x_1 and x_2 , computed along dim . x_1 and x_2 must be broadcastable to a common shape. dim refers to the dimension in this common shape. Dimension dim of the output is squeezed (see `torch.squeeze()`), resulting in the output tensor having 1 fewer dimension. Supports type promotion. x_1 (Tensor) – First input. x_2 (Tensor) – Second input. dim (int, optional) – Dimension along which cosine similarity is computed. Default: 1

Function Signature

```
torch.nn.functional.cosine_similarity(x1, x2, dim=1, eps=1e-8) → Tensor#
```

Parameters

Code Examples

```
>>> input1 = torch.randn(100, 128)
>>> input2 = torch.randn(100, 128)
>>> output = F.cosine_similarity(input1, input2)
>>> print(output)
```

Detailed Documentation

Documentation

Returns cosine similarity between x_1 and x_2 , computed along dim . x_1 and x_2 must be broadcastable to a common shape. dim refers to the dimension in this common shape. Dimension dim of the output is squeezed (see `torch.squeeze()`), resulting in the output tensor having 1 fewer dimension.

Supports type promotion.

x_1 (Tensor) – First input.

x_2 (Tensor) – Second input.

dim (int, optional) – Dimension along which cosine similarity is computed. Default: 1

eps (float, optional) – Small value to avoid division by zero. Default: $1\text{e-}8$

